

method	main effect	zero / pole side effect	best use case
PD series	ζ up, over- shoot down	adds one zero	shape tran- sient response
velocity feedback	ζ up, ω_n fixed	no extra zero	improve damping with cleaner structure
input feedforward	reduce track- ing error	needs model matching	raise steady- state accuracy
disturbance compensation	cancel load effect	sensitive to disturbance model	disturbance rejection